Project Name: Lab Consumable Expiry Tracker

Team Members:

1. Hary Zulfa Supriyanto
2. Mahindra Irvan Saputra

Problem (The "Why")

- **Current Issue:** Laboratory reagents and consumables generally have fixed shelf lives, and many require refrigerated or otherwise controlled storage. In many labs, tracking which lots are still valid, how much stock remains, and which lots are approaching expiry is done manually on spreadsheets, whiteboards, or by memory. There is no automated mechanism that prevents an expired or near-empty reagent from being pulled and used in an experiment or production run.
- **Impact:** When an expired reagent is used without anyone noticing, the result is a silent quality defect an experiment result or QC test that looks valid but isn't, and the mistake is often only discovered after time and materials have already been spent, or worse, after a faulty batch has moved downstream. Lab staff carry the burden of manually cross-checking expiry dates before every use, and procurement teams have no reliable, real-time signal for when to reorder before a critical reagent actually runs out. This is a common pain point across laboratory and manufacturing environments that handle consumables with limited shelf life — not tied to any single organization's specific inventory.

Proposed Solution

- **Our Approach:** The system tracks every reagent/consumable lot registered in the lab, including its current quantity, expiry date, and configured minimum stock level per item type. Each time a lot is consumed (partially or fully) in a job, its remaining quantity is updated. The system continuously evaluates every lot's status Available, Expiring Soon, Low Stock, or

Expired/Blocked and automatically blocks any lot that has expired from being selected for use in a new job, even if quantity remains. When multiple valid lots of the same reagent exist, the system recommends the lot with the nearest expiry date first (First-Expired-First-Out), so stock is never left to expire unused.

- **Key Value:** The entire expiry and stock lifecycle from quantity deduction, to expiry evaluation, to status transition and FEFO lot selection is pure business logic with no UI or database dependency, making it testable at 100% coverage using .NET 8 TimeProvider abstraction. This means the most critical behaviour of the system (blocking an expired lot and recommending the right lot to use next) can be verified precisely and automatically, giving the team and lab management confidence that no expired reagent can slip into a job undetected.

Core Features

- **Feature 1:** Lot Quantity Tracker — records each reagent/consumable lot's remaining quantity, deducting it every time the lot is consumed against a job, and comparing it against a configurable minimum stock threshold defined per item type.
- **Feature 2:** Expiry Status Engine — evaluates every lot's current status (Available, Expiring Soon, Low Stock, or Blocked/Expired) based on its expiry date and remaining quantity, and automatically excludes any expired lot from being assigned to a new job.
- **Feature 3:** FEFO Lot Recommendation & Consumption Log — when a reagent is needed for a job, the system recommends the valid lot with the nearest expiry date first, records every consumption event against that lot, and produces a full auditable history of stock usage and disposals.

High-Level Architecture & Quality Assurance

- **Architecture Pattern:** Clean Architecture Domain and Application layers are fully independent of infrastructure, ensuring all expiry, stock, and lot-selection logic can be unit-tested in complete isolation.
- **Design Patterns Used:** Repository Pattern (lot and consumption data access), Dependency Injection, and .NET 8 TimeProvider abstraction for deterministic, expiry-date-based testing without mocking the system clock directly.

- **Code Quality Goal:** We commit to maintaining a green CI pipeline (build + test on every push), writing unit tests tracked via a coverage tool (e.g. Coverlet/Cobertura), enforcing zero compiler warnings via TreatWarningsAsErrors, and keeping static analysis warnings at the lowest achievable level (Target: 0).

Identified Challenges & Risks

- **Risk 1:** Accurately modelling the lot status state machine, particularly edge cases such as a lot that expires mid-job, a partially consumed lot that later expires, or two lots with the same expiry date, so that no expired or invalid lot can slip through to production use.
- **Risk 2:** Designing expiry thresholds and minimum stock rules to be configurable per reagent/item type without hardcoding, so the system remains maintainable as new reagent types are introduced to the lab, a direct application of the Open/Closed Principle.